

Midhun Chakkaravarthy K

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EXPERIENCE

- Tata Elxsi** Bengaluru, India
Software Engineer June 2025 – Present
 - Own **Content Catalog and Metadata Services** built with **Node.js, Express.js, REST APIs, and Microsoft SQL**, enabling ingestion, normalization, and retrieval of **thousands of content assets** across multi-device OTT platforms.
 - Integrate **VAST/VMAP-based AdTech Workflows** with backend services to support server-side and client-side ad delivery, content monetization, and seamless playback experiences.
 - Drive **User Analytics and Personalization Services**, processing **thousands of playback and session events** to power real-time analytics, content recommendations, and audience engagement insights.
 - Maintain and enhance the **beIN Sports Editorial Content Platform** across **Web, Android, and iOS**, supporting multilingual content delivery in **4 languages** across multiple global regions.
 - Contribute scalable backend features using **Node.js, Express.js, Microsoft SQL, REST APIs, and Microservices Architecture**, ensuring reliable content delivery during high-traffic sporting events.
- Sayvai Software** Remote
Full Stack Developer June 2024 – May 2025
 - Architected a **Voice AI Agent Management Platform** using **Next.js, React, FastAPI, PostgreSQL, STT, TTS, and LLMs**, enabling centralized creation and deployment of conversational voice agents.
 - Designed **Multi-Agent Administration Dashboards** supporting lifecycle management, deployment configuration, monitoring, and management of multiple AI agents through a unified interface.
 - Created a **CMS and Website Builder Platform** using **Next.js, FastAPI, Dynamic Routing, and Reusable Components**, delivering **6+ reusable modules** for page composition and configurable chat and voice integrations.
 - Developed a **Python/Flask-based Parametric Pattern Processing System** converting GarmentCode patterns into **SVG, PNG, PDF, and DXF** formats while preserving production-grade geometric accuracy.
 - Established a **3D Garment Visualization Platform** using **Meshcapade API** and adopted **Microservices Architecture** with Flask, Supabase, SQLAlchemy, and Alembic to support scalable rendering and fit-validation workflows.
- Smartail Private Limited** Chennai, India
Product Engineer Intern December 2023 – May 2024
 - Redesigned the **admin portal** for **70+ schools** and **65,000+ students**, implementing class, subject, syllabus, and publication modules using **Angular** and **Spring Boot**.
 - Built features and resolved major issues across the stack using **Angular, Spring Boot, and Flask**; delivered **10+ UI components** with corresponding APIs and improved application stability.
 - Utilized **Docker** for consistent environments and **Kafka** for real-time data flows, enhancing system reliability.
- Smartail Private Limited** Chennai, India
Product Engineer Intern June 2022 – December 2022
 - Built an **image-extraction workflow** for subjective and objective questions, integrating **MathLive Keyboard** and **KaTeX** for accurate math rendering and extraction from images.
 - Developed multiple UI components and full pages using **Angular**, and integrated APIs with a **Spring Boot** backend to ensure consistent data flow and system functionality.

SKILLS SUMMARY

- Languages:** C, Java, Python, JavaScript, TypeScript, SQL.
- Frameworks & Libraries:** Angular, React, React Hooks, Redux, Next.js, Node.js, Express.js, FastAPI, Flask, Spring Boot, Tailwind CSS, LangChain, SQLAlchemy, Alembic.
- Databases:** MongoDB, PostgreSQL, MySQL, Microsoft SQL Server, Supabase, Vector Databases.
- Tools:** AWS, Azure, Azure DevOps, Docker, Kubernetes, Git, Kafka, Postman, WebSockets, CI/CD.
- Core:** MERN Stack, REST API Design, Microservices Architecture, Distributed Systems, Event-Driven Architecture, System Design.

EDUCATION

- Coimbatore Institute of Technology** Coimbatore, India
Master of Science in Software Systems, Cumulative GPA: 8.43/10 July 2019 – May 2024